BrightLearn Data Analytics

# Research Assignment 1

## Foundations of Data Analytics & Data Science

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| Course | Data Analytics & Data Science |
| Assignment No. | Research Assignment 1 |
| Target Audience | Beginner Students |
| Due Date | To be confirmed by instructor |
| Format | Handwritten answers — scan and email your completed work |

## Instructions

Read each question carefully and research the answer before writing.

- All answers must be handwritten — typed submissions will not be accepted.
- Write in full sentences and use your own words.
- Include diagrams or examples where relevant — this will earn you bonus marks.
- Cite your sources at the bottom of each section.
- Scan your completed assignment and submit it before the due date.

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## Section A: Data Fundamentals

**Research Tip:** *Start with official documentation from platforms like IBM, Google Cloud, AWS, or Microsoft Azure — they offer excellent beginner-friendly explanations.*

**1.** What is a Database? Define it in your own words and explain why databases are essential in the modern world. Provide two real-world examples of where databases are used.

A database is an organized collection of data that is stored and managed so that it can be easily accessed and updated. Databases are essential because they allow for secure and quick retrieval

~~of~~ as well as storage of large amounts of information or data. Examples include banking systems that store customer accounts and e-commerce platforms that store product and customer data.

**2.** What is a Data Warehouse? How does it differ from a regular database? Why would a company use a data warehouse?

A data warehouse is a system that stores large amounts of current and historical data for analysis and reporting. A regular database is used for daily operations, while a data warehouse is used for analysis. Companies use data warehouses to make better decisions based on trends over time.

**3.** What is a Data Lake? Explain the concept and describe how it differs from a data warehouse. When would a business prefer a data lake over a data warehouse?

A data Lake is a storage system that holds raw data in its original format. It differs from a data warehouse because it does not require data to be structured before it is stored. Businesses would prefer Data lake when they need to store large volumes of diverse data for future analysis.

**4.** What is a Data Mart? How does it relate to a data warehouse and who typically uses it?

A data Mart is a smaller section of a data warehouse focused on a specific business area like finance or sales. It is used by specific teams within an organization for focused analysis.

**5.** Explain the difference between Structured, Semi-Structured, and Unstructured Data. Provide one example of each.

Structured data is organized in tables with rows and columns such as a customer database. Semi structured data has some organisation but not fixed tables, such as JSON files. Unstructured data has no clear structure, such as images or videos.

**6.** What is Data Modeling? Why is it important when designing a system that works with data? Describe at least two types of data models.

Data Modeling is the process of designing how data is stored and related in a system. It is important because it ensures data is organised, consistent and easy to use. Examples include Conceptual models and logical models.

**7.** What is Data Mining? How is it used in business? Provide two real-world examples of data mining in action.

Data Mining is the process of analysing large datasets to find patterns and insights. Businesses use it to make informed decisions. ~~For~~ For example, retailers use it to recommend products to customers, and banks use it to detect fraud

**8.** What is a Database Management System (DBMS)? Name and briefly describe four popular DBMS platforms (e.g., MySQL, PostgreSQL, Oracle, MongoDB) and explain a suitable use case for each.

A DBMS is a software that manages databases and allows users to create, retrieve, update and delete data. MySQL is used for web applications, PostgreSQL is used for complex data analysis, Oracle is used in large enterprises and MongoDB is used for flexible data like documents.

**9.** What is ETL (Extract, Transform, Load)? Explain each step and describe why this process is important in data analytics.

ETL stands for Extract, Transform and Load. Extract means collecting data from sources; Transform means cleaning and preparing the data. Load means storing it in a system like a data warehouse. It is an important process because it ensures data is accurate and ready for analysis.

**10.** What are common data storage formats used in analytics (e.g., CSV, JSON, Parquet, Avro)? Briefly describe each and state when you would use one over another.

CSV is a simple text format used for small datasets- CSV stands for Comma Separated Values

JSON is used for data exchange in APIs- It stands for JavaScript Object Notation

Parquet is a column based format used for big data and fast queries.

Avro is used for storing large data with a defined schema, often in streaming systems

## Section B: Artificial Intelligence & Machine Learning

**Research Tip:** *Explore resources like Google's Machine Learning Crash Course, IBM's AI explanations, or OpenAI's blog to build a strong foundational understanding.*

**11.** What is Artificial Intelligence (AI)? Define it and explain the difference between Narrow AI (Weak AI) and General AI (Strong AI). Provide one example of each.

AI is the ability of machines to perform tasks that require human intelligence. Narrow AI is designed for specific tasks like voice assistants. General AI would be able to perform any intellectual task like a human. An example of Narrow AI is a chatbot. True General AI does not currently exist.

**12.** What is Machine Learning (ML)? How does it differ from traditional programming? Describe the three main types of Machine Learning: Supervised, Unsupervised, and Reinforcement Learning.

ML is a method where machines learn from data instead of being explicitly programmed. Traditional programming uses fixed rules while ML learns patterns from data. Supervised learning uses labeled data, unsupervised ~~data~~ learning finds patterns in unlabeled data. Reinforcement learning learns through rewards and penalties.

**13.** What is Deep Learning? How does it relate to Machine Learning and Artificial Intelligence? What are Neural Networks and how do they work?

Deep learning is a type of machine learning that uses neural networks with many layers. It is a subset of ML, which is part of AI. Neural Networks are systems that mimic the human brain and learns patterns by adjusting weights based on input data.

**14.** What is Natural Language Processing (NLP)? Describe what it is and provide three real-world applications of NLP that you use in everyday life.

NLP is the ability of machines to understand and process human language. Examples include chatbots, voice

assistants and email spam filters.

**15.** What is a Large Language Model (LLM)? How is it different from a standard Machine Learning model? Name two well-known LLMs and explain what they are used for.

An LLM is an AI model trained on large amounts of text to understand and generate human language. It differs from standard models because it handles complex language tasks. Examples include GPT models used for conversation and text generation and Claude used for writing and analysis.

**16.** What is Generative AI? How does it work at a high level? Name three tools that use Generative AI and describe what each one does.

Generative AI creates new content like text, images or code. It works by learning patterns from data and generating similar outputs. Examples include ChatGPT for text, DALL E for images, and Github Copilot for code.

**17.** What is a Training Dataset? Why is the quality of training data so important when building an AI model? What could go wrong if the training data is biased or incorrect?

A training Dataset is the data used to teach an AI model. High quality data is important because it affects the model's accuracy. If the model is incorrect or biased, the model can produce unfair or wrong results

**18.** What is Computer Vision? How do machines 'see' and interpret images? Provide two real-world use cases of computer vision.

Computer Vision is a field where machines interpret images and videos. Machines use algorithms to detect patterns in visual data. Examples include facial recognition and self driving cars detecting objects and obstacles.

## Section C: APIs, Tools & Developer Concepts

**Research Tip:** *Search for 'What is an API for beginners' on YouTube or read documentation from platforms like Postman, RapidAPI, or Anthropic.*

**19.** What is an API (Application Programming Interface)? Explain what it is using a simple real-world analogy. Why are APIs important in modern software and data systems?

An API is a tool that allows different software systems to communicate. It is like a waiter taking orders between a customer and a kitchen. APIs are important because they allow systems to share data and services easily

**20.** What is an API Key? What is its purpose? How should it be stored and protected, and what could happen if it is exposed publicly?

An API key is a unique code used to identify and ~~auuthes~~ authenticate a user or application. It should be stored securely and not shared publicly. If exposed, others can misuse the service and cause security issues.

**21.** What is a REST API? What does REST stand for and what are the main HTTP methods (GET, POST, PUT, DELETE) used in REST APIs? Briefly describe each method.

REST stands for Representational State Transfer. GET retrieves data. POST sends new data, PUT updates existing data and DELETE removes data

**22.** What is JSON (JavaScript Object Notation)? Why is it commonly used when working with APIs and data exchange? Write a simple example of a JSON object.

JSON is a lightweight format for storing and exchanging data. It is widely used because it is simple and easy to read.

Example :

{ "name" : "John",
"age" : 25 }

**23.** What is Prompt Engineering? Why is it an important skill when working with AI tools like ChatGPT or Claude? Provide two examples of a poorly written prompt and an improved version of each.

Prompt engineering is the practice of writing clear instructions for AI tools. It is important because better prompts lead to better outputs. Examples of poor prompt : write code. Improved: Write a python function to calculate average of a list. Poor prompt: Tell me about data. Improved: Explain data analytics in simple terms with examples.

**24.** What is a MAP function? Explain what it does in programming and data processing. Provide a simple example in either Python or JavaScript.

A MAP function applies a function to each item in a list.
Example in Python:

```
numbers = [1, 2, 3]
result = list(map(lambda x: x*2, numbers))
```

**25.** What is Cloud Computing? Name the three main cloud service models (IaaS, PaaS, SaaS) and briefly explain each. Give one example of a product for each model.

Cloud computing is the delivery of computing services over the Internet. IaaS provides infrastructure like servers, PaaS provides platforms for development. and SaaS provides ready to use software. Examples include AWS for IaaS, Google App Engine for PaaS and Google drive for SaaS

**26.** What is a Version Control System? What is Git and why is it widely used by data analysts and developers? Briefly explain what a 'commit' and a 'branch' are.

A version control system tracks changes in code. Git is widely used because it allows collaboration and history tracking. A commit is a saved change and a branch is a separate version of the code for development.

**27.** What is a Jupyter Notebook? What programming language is it most commonly used with? Why is it popular among Data Analysts and Data Scientists?

A Jupyter Notebook is an interactive environment for writing and running code. It is most commonly used with Python. It is popular because it allows combining code, text and visuals in one place.

**28.** What is Python? Why is it considered one of the most popular programming languages for Data Analytics and AI? Name four Python libraries commonly used in data work and briefly describe each.

Python is a high-level, general-purpose programming language known for its readability, simplicity and versatility. It is popular in data analytics because it has many libraries. Pandas is used for data analysis, Numpy for numerical operations, Matplotlib for visualisation and Scikit learn for Machine learning.

## Section D: Data Ethics, Governance & Careers

**Research Tip:** *Explore the World Economic Forum, GDPR documentation, and LinkedIn Learning for content on data ethics and career pathways in data.*

**29.** What is Data Privacy? Why is it important? Research the GDPR (General Data Protection Regulation) and explain what it is and which countries it applies to.

Data ~~protection~~ privacy is the protection of personal data. It is important to prevent misuse of information. GDPR is a law in the European Union that protects personal data and applies to organisations that handle EU citizens' data.

**30.** What is Data Ethics? Provide three examples of unethical use of data or AI, and explain what makes each example problematic.

Data Ethics refer to the responsible use of data. Examples of unethical data usage include using personal data without consent, biased AI decisions, selling user data without transparency. These are problematic because they harm trust and fairness.

**31.** What is Data Governance? Define it and explain the key components of a good data governance framework.

Data governance is the management of data availability, quality and security. Key components include data quality, data security, data policies and clear roles and responsibilities.

**32.** What is Bias in AI? Explain what it means for an AI model to be biased. Provide a real-world example of AI bias and describe the potential consequences.

Bias in AI means the model produces unfair results due to biased data. For example, a hiring system favoring certain groups of people can lead to discrimination and unfair outcomes.

**33.** What are the key differences between a Data Analyst, a Data Scientist, a Data Engineer, and an AI Engineer? Create a simple table comparing their roles, skills, and tools.

| Role | Focus | Skills | Tools |
|---|---|---|---|
| Data Analyst | Analyse data & create reports | Excel, SQL, visualisation | Power BI, Tableau |
| Data Scientist | Build models and predictions | statistics, ML, coding | Python, R |
| Data Engineer | Build data systems & pipelines | Databases, ETL, coding | SQL, Spark |
| AI Engineer | Develop AI systems | ML, Deep learning | TensorFlow, PyTorch |

**34.** What is Business Intelligence (BI)? Name two popular BI tools (e.g., Power BI, Tableau) and explain how they help businesses make data-driven decisions.

Business Intelligence is the use of data to support decision making Tools like PowerBI and Tableau help by creating dashboards and visual reports

## Section E: Personal Reflection

Answer the following reflection questions thoughtfully and honestly. There is no right or wrong answer — we want to know what you think!

**35.** Which concept from this assignment did you find most interesting or surprising? Explain why.

The most interesting concept was Machine learning because it shows how systems can learn data and improve over time

**36.** Which concept did you find most difficult to understand? What resources did you use to help you understand it better?

Deep learning was the most difficult. I used online tutorials

and videos to understand neural networks better.

**37.** How do you see Data and AI affecting your future career or everyday life? Be specific with examples.

Data and AI will impact my career by helping me make better decisions and automate tasks. For example, I can use data analysis to improve business performance and AI to increase productivity

## References

List all the sources you used to complete this assignment. Use the format below:

Author/Organisation. (Year). Title of article or webpage. Website Name. URL


1. IBM (2023). What is data analysis. IBM Website.
2. Google Cloud (2023) Datawarehouse explained. Google Cloud Website.
3. AWS. (2023) What is data lake. AWS website.
4. Microsoft (2023) Introduction to AI. Microsoft Learn
5. OpenAI (2023) AI and language models. OpenAI Website.
6. Oracle. (2023) Database Concepts. Oracle Website
7. Python.org (2023). Python documentation. Python Website.
8. GDPR EU (2023). General Data Protection Regulation. EU website


Good luck! We believe in you.
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